Retinal Image Analysis for Neurological Disease Detection: A Biomarker and Deep Learning Pipeline

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Abstract

Neurodegenerative diseases such as Parkinson's disease (PD) represent a major global health challenge, affecting millions of individuals worldwide. Early detection of these conditions remains difficult because clinical diagnosis typically occurs only after significant neuronal damage has already taken place. Recent studies suggest that **the retina may serve as a potential window into neurological health due to its shared embryological origin with the central nervous system.**

This project investigates whether retinal fundus images can be used to extract quantitative vascular biomarkers associated with neurological disorders. A computational pipeline was developed to perform retinal vessel segmentation using deep learning and subsequently extract interpretable vascular biomarkers including vessel density, tortuosity, and fractal dimension. The segmentation model was trained using publicly available retinal vessel datasets including DRIVE, STARE, and CHASE_DB1. Experimental results demonstrate strong segmentation performance with **an accuracy of 0.9408 and ROC-AUC of 0.9525 on the DRIVE dataset.**

In addition to biomarker extraction, a papilledema detection experiment was conducted as a clinically motivated proof-of-concept for retinal-based neurological screening. Using EfficientNet-B0, the model achieved 99% accuracy on the Kaggle training dataset and **92% accuracy on an external Bangladesh clinical dataset.** Grad-CAM interpretability analysis confirmed that the model attends to the optic disc region in 70% of test cases, consistent with the clinical diagnostic criteria for papilledema.

Following this, a transition was made from global vascular biomarkers to localised optic disc feature extraction, yielding five statistically significant features ($p < 0.001$). A comparison of Random Forest and Logistic Regression classifiers demonstrated near-identical F1 scores (0.700 vs. 0.710), while a Hybrid Random Forest combining disc features with CNN confidence achieved $\text{F1} = 0.965$. Multicollinearity analysis and PCA-based feature decorrelation were subsequently conducted, revealing that PCA with all five components improved both RF F1 to 0.762 and LR F1 to 0.757, confirming that multicollinearity was a contributing factor to LR underperformance on the original feature set.

These results demonstrate that retinal image analysis can successfully detect both vascular and structural retinal abnormalities associated with neurological disease. The developed pipeline provides a foundation for future work aimed at detecting Parkinson's disease using retinal biomarkers.

Contents

1 Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by the degeneration of dopaminergic neurons in the substantia nigra. The disease affects more than 10 million individuals globally and leads to motor symptoms including tremor, rigidity, and impaired balance, as well as non-motor symptoms such as cognitive decline.

One of the major challenges in Parkinson's disease management is the lack of reliable early diagnostic tools. Clinical diagnosis is typically based on motor symptoms, which appear only after substantial neuronal loss has already occurred. Therefore, identifying non-invasive biomarkers capable of detecting early neurodegeneration has become an important research objective.

Recent advances in ophthalmology and neuroscience have highlighted the retina as a potential source of neurological biomarkers. The retina is considered an extension of the central nervous system and shares similar microvascular and neuronal structures with the brain. As a result, retinal abnormalities may reflect neurodegenerative processes occurring in the brain.

Retinal fundus imaging provides a non-invasive and widely accessible method for visualizing retinal structures and blood vessels. Several studies have reported **retinal vascular changes associated with neurological diseases, including reduced vessel density, altered vascular geometry, and changes in vascular complexity.**

This project explores whether computational analysis of retinal fundus images can reveal biomarkers relevant to neurological disorders. Specifically, the research focuses on developing a deep learning based pipeline for retinal vessel segmentation and quantitative biomarker extraction.

Because publicly available retinal datasets labeled specifically for Parkinson's disease are limited, the study also includes an intermediate experiment on papilledema detection. Papilledema, which refers to swelling of the optic disc caused by increased intracranial pressure, represents a visible retinal manifestation of neurological pathology and is associated with several neurological conditions involving elevated intracranial pressure. **Detecting papilledema from retinal images provides a proof-of-concept for retinal-based neurological screening.** Furthermore, the optic disc features developed for papilledema detection are structurally related to the disc-level biomarkers implicated in Parkinson's disease, including optic disc pallor and peripapillary RNFL thinning, making this intermediate task scientifically motivated rather than incidental.

The long-term objective of this research is to develop an automated retinal image analysis framework capable of detecting neurological disease biomarkers using artificial intelligence.

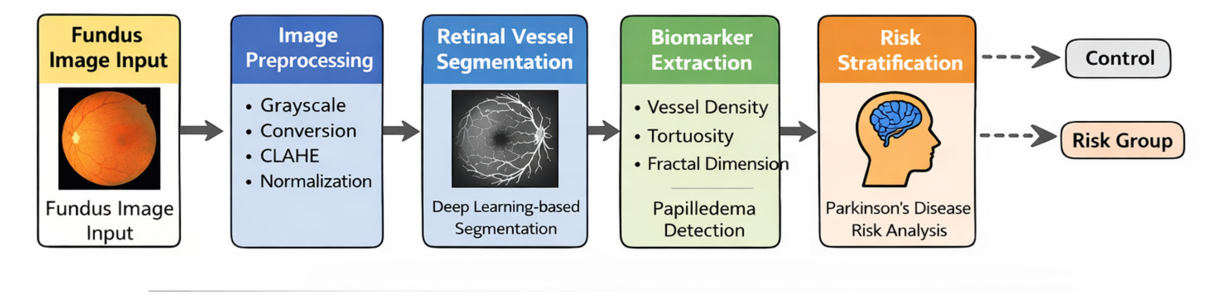


Figure 1: Overview of the proposed retinal biomarker extraction and neurological risk assessment pipeline.

2 Research Objectives

The primary objective of this research is to investigate whether retinal fundus images can provide measurable biomarkers associated with neurodegenerative diseases such as Parkinson's disease. Specifically, the research focuses on the following aims:

1. Developing a deep learning based retinal vessel segmentation model capable of accurately extracting retinal vascular structures.
2. Computing quantitative vascular biomarkers including vessel density, tortuosity, and fractal dimension from segmented vessel masks.
3. Evaluating whether these biomarkers produce physiologically meaningful measurements consistent with findings reported in ophthalmology literature.
4. Investigating whether retinal images can reveal neurological abnormalities through the detection of papilledema using deep learning classification models, as a proof-of-concept for retinal-based neurological screening.
5. Validating cross-dataset generalisation of the papilledema classifier on an external clinical fundus dataset.
6. Assessing model interpretability using Grad-CAM to confirm that the classifier attends to clinically relevant anatomical regions.
7. Extracting localised optic disc features directly capturing the clinical signs of papilledema, and evaluating their discriminative power.
8. Comparing the performance of a black-box classifier (Random Forest) with an explainable classifier (Logistic Regression), and investigating the effect of combining disc features with CNN confidence in a Hybrid model.
9. Investigating multicollinearity within the disc feature set and applying PCA-based decorrelation to assess its effect on classifier stability and performance.
10. Establishing a computational retinal biomarker analysis pipeline extensible to Parkinson's disease datasets when labelled retinal images become available.

3 Retina as a Biomarker for Neurological Health

3.1 Retina as a Window into Neurological Health

The retina is increasingly being studied as a potential biomarker source for neurological diseases. Because the retina originates from the same embryological tissue as the brain, retinal structures and microvasculature may reflect neurodegenerative processes occurring in the central nervous system.

Previous studies have reported structural retinal changes in patients with Parkinson's disease, **including thinning of the retinal nerve fiber layer (RNFL) and ganglion cell-inner plexiform layer (GCIPL)**. These structural changes are believed to mirror neuronal degeneration occurring in the brain [2].

3.2 Retinal Vascular Biomarkers

In addition to structural retinal changes, alterations in retinal microvasculature have also been associated with neurological disorders. Several vascular biomarkers have been proposed in the literature:

Table 1: Common retinal vascular biomarkers and their clinical relevance

Biomarker	**Description**	**Clinical Relevance**
Vessel Density	Proportion of vessel pixels in retinal region	Reduced perfusion in neurodegeneration
Tortuosity	Measure of vessel curvature	Indicates vascular abnormalities
Fractal Dimension	Complexity of vascular branching	Reflects microvascular integrity

These biomarkers can be computed from retinal vessel segmentation masks, making vessel segmentation an essential preprocessing step for retinal biomarker analysis.

3.3 Deep Learning for Retinal Vessel Segmentation

Retinal vessel segmentation is a challenging problem because retinal blood vessels vary significantly in width, orientation, and contrast. Traditional image processing approaches often struggle to detect thin vessels and may produce fragmented vessel structures.

Deep learning approaches, particularly convolutional neural networks (CNNs), have significantly improved segmentation performance. The U-Net architecture has become widely used for biomedical image segmentation due to its encoder-decoder structure with skip connections, which preserves spatial information while capturing contextual features [10]. The retinal vascular biomarkers described above, such as vessel density, tortuosity, and fractal dimension, cannot be directly computed from raw fundus images. Accurate extraction of these biomarkers requires precise segmentation of the retinal vasculature. Therefore, retinal vessel segmentation serves as a critical intermediate step that enables quantitative analysis of microvascular structures associated with neurological disorders like Parkinson's disease. In this context, deep learning-based segmentation methods are employed to generate vessel masks from which these biomarkers can be computed and further used for disease risk assessment.

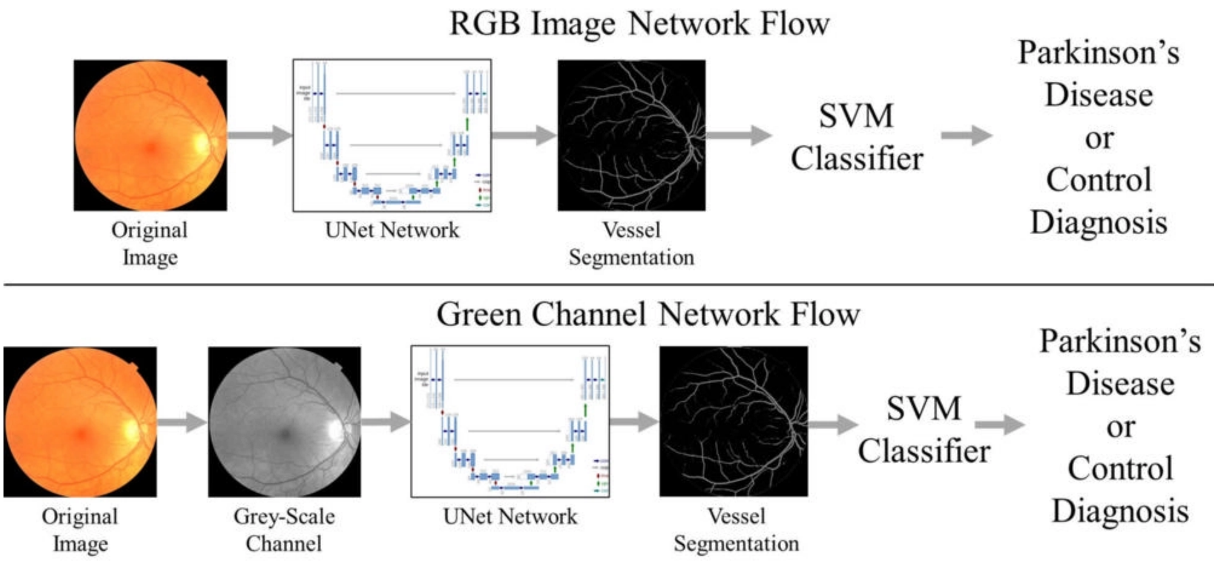


Figure 2: Overview of the retinal extraction pipeline developed in this project (adapted from [1]).

4 Dataset and Problem Formulation

One of the primary limitations encountered during this research was the lack of openly available retinal fundus datasets labeled specifically for Parkinson's disease. Large biomedical datasets such as UK Biobank and the INSIGHT datalink contain retinal imaging data, but require formal access applications and are not directly downloadable for research use.

The following datasets were used for vessel segmentation training and evaluation. These datasets contain manually annotated vessel masks, allowing supervised training of segmentation models.

Table 2: Retinal vessel segmentation datasets

Dataset	**Images**	**Resolution**	**Purpose**
DRIVE	40	768×584	Training and evaluation
STARE	20	605×700	Cross-dataset validation
CHASE_DB1	28	High resolution	Generalisation testing

Due to this limitation, the current study focuses on developing a retinal biomarker extraction pipeline using publicly available vessel segmentation datasets. A two-stage strategy was adopted: vessel segmentation models were trained using established retinal vessel datasets, and the extracted vascular biomarkers can later be applied to neurological disease datasets when they become available.

Figure 3: Example retinal fundus image showing the optic disc and vascular network.

5 Retinal Vessel Segmentation Methodology

Retinal vessel segmentation forms the foundation of the biomarker extraction pipeline. Accurate segmentation of retinal blood vessels enables the computation of vascular metrics that may serve as biomarkers for neurological disease.

5.1 Image Preprocessing

Fundus images exhibit variability in illumination, contrast, and background intensity. Preprocessing was therefore applied to enhance vessel visibility and standardize image characteristics before segmentation. The preprocessing pipeline consisted of the following steps:

- Conversion of RGB fundus images to grayscale to reduce channel redundancy.
- Dataset normalization using the mean and standard deviation computed across all training images.
- Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance vessel contrast.
- Gamma correction to improve brightness distribution.
- Normalization of pixel values to the range $[0, 1]$ for neural network input.

These preprocessing steps improve the visibility of thin vessels and reduce noise introduced by uneven illumination.

5.2 Patch-Based Training Strategy

Full-resolution fundus images are computationally expensive to process directly. To address this, images were divided into overlapping patches of size 48×48 pixels. The patch-based training strategy provides several advantages: it increases the effective number of training samples, improves detection of fine vascular structures, and reduces GPU memory requirements. Overlapping patches ensure that vessel continuity is preserved when predictions are recombined into full-image probability maps.

5.3 Segmentation Network Architecture

A U-Net based architecture was implemented for vessel segmentation. U-Net is widely used in biomedical image segmentation because it combines local feature extraction with contextual understanding through its encoder-decoder structure. The encoder progressively reduces spatial resolution while extracting high-level features, while the decoder restores spatial resolution using skip connections from corresponding encoder layers.

Table 3: Segmentation model configuration

Parameter	**Value**
Architecture	U-Net
Patch size	48×48
Loss function	Cross Entropy
Optimiser	SGD
Learning rate	0.01
Dropout rate	0.2

The use of skip connections enables the network to retain fine spatial details that are essential for detecting thin retinal vessels.

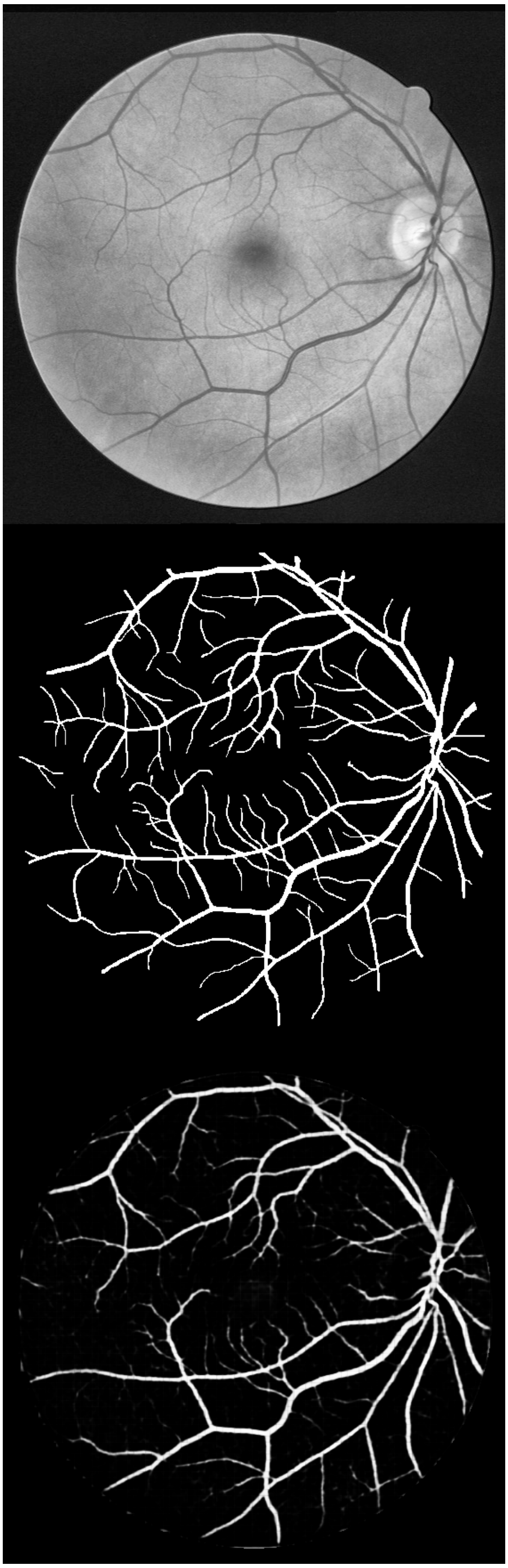


Figure 4: Example retinal vessel segmentation result showing the original fundus image, ground truth vessel mask, and predicted vessel mask.

6 Segmentation Results

The vessel segmentation model was evaluated on the DRIVE dataset using several standard performance metrics in biomedical image segmentation, including the ROC-AUC, Precision–Recall AUC, Jaccard similarity coefficient, F1 score, accuracy, sensitivity, specificity, and precision.

Table 4: Retinal vessel segmentation performance on the DRIVE dataset

Metric	**Value**
ROC-AUC	0.9525
Precision–Recall AUC	0.8463
Jaccard Similarity (IoU)	0.6217
F1 Score	0.7667
Accuracy	0.9408
Sensitivity (Recall)	0.7638
Specificity	0.9667
Precision	0.7697

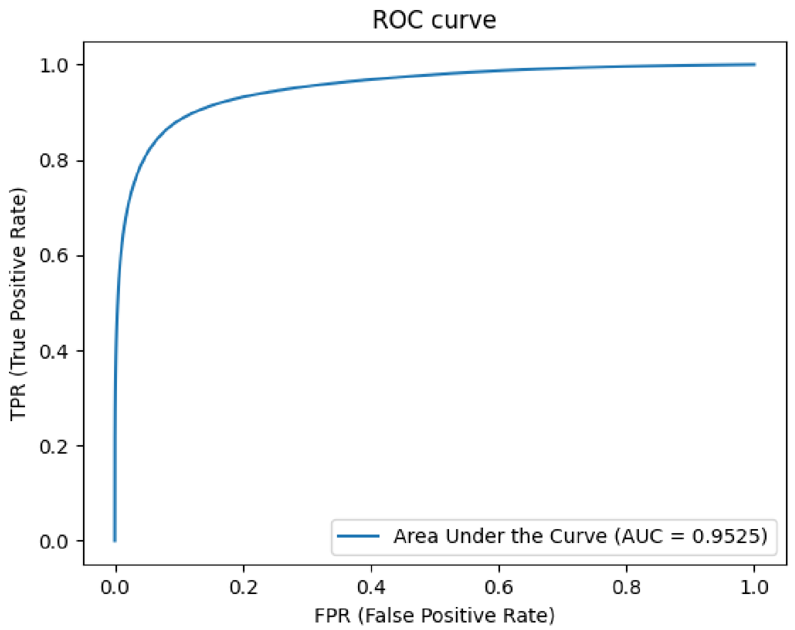


Figure 5: ROC curve for the retinal vessel segmentation model on the DRIVE test set.

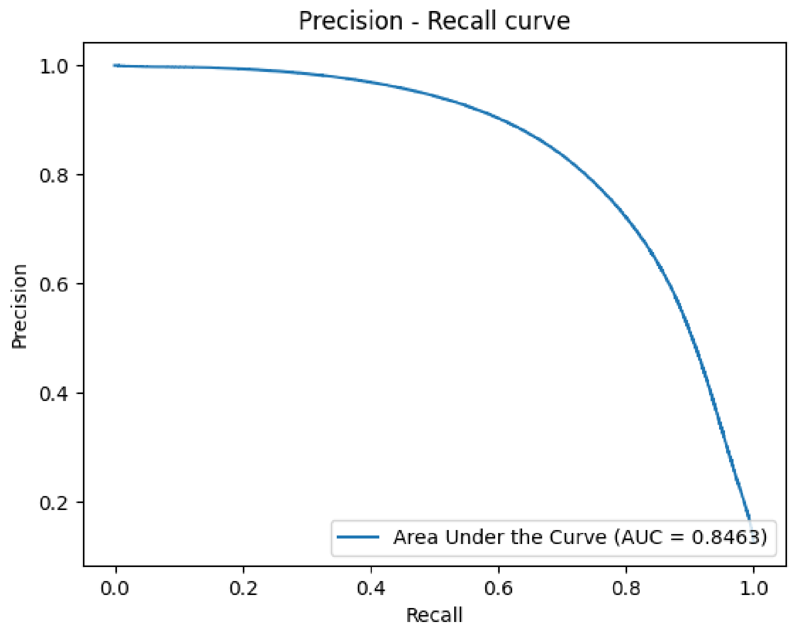


Figure 6: Precision-Recall curve for the retinal vessel segmentation model.

The high ROC-AUC score of 0.9525 indicates that the segmentation model effectively distinguishes vessel pixels from background pixels across varying classification thresholds. The Precision–Recall AUC of 0.8463 further demonstrates strong performance in detecting vessel structures despite the class imbalance typically present in retinal images.

The Jaccard similarity coefficient of 0.6217 and F1 score of 0.7667 indicate that the predicted vessel masks closely match the manually annotated ground truth masks, measuring the overlap between predicted and true vessel regions. The high specificity of 0.9667 indicates that the model rarely misclassifies background pixels as vessels, which is particularly important for downstream biomarker computation since false positive vessel detections can distort measurements such as vessel density and fractal dimension.

7 Retinal Biomarker Extraction

After vessel segmentation, quantitative vascular biomarkers were computed from the resulting vessel masks. These biomarkers provide interpretable metrics describing the structure of the retinal vascular network.

7.1 Vessel Density

Vessel density measures the proportion of vessel pixels relative to the total retinal area. Reduced vessel density has been reported in several neurological disorders and may reflect compromised microvascular perfusion [7].

7.2 Vessel Tortuosity

Tortuosity measures the curvature of blood vessels. It is computed by comparing the path length of a vessel segment, extracted from the skeletonised vessel mask, to the straight-line

Euclidean distance between its endpoints.

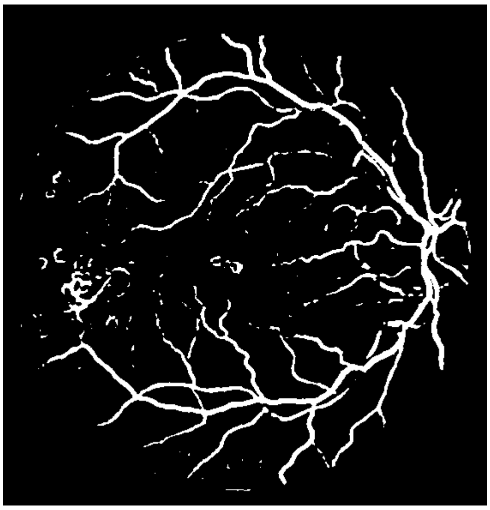


Figure 7: Example vessel skeleton representation used for biomarker computation including tortuosity and fractal dimension.

7.3 Fractal Dimension

Fractal dimension quantifies the complexity of vascular branching patterns. It is computed using box-counting analysis on the binarised vessel mask [5].

Table 5: Extracted retinal biomarkers on the DRIVE dataset

Biomarker	Description	Observed Value
Vessel Density	Ratio of vessel pixels	0.103 ± 0.015
Tortuosity	Vessel curvature measure	2.38 ± 0.30
Fractal Dimension	Vascular branching complexity	1.394 ± 0.033

These biomarker values fall within ranges reported in the ophthalmology literature for healthy retinal vasculature, indicating that the segmentation pipeline produces physiologically meaningful results.

7.4 Vascular Biomarkers on the Bangladesh Dataset

The DRIVE-trained U-Net was subsequently applied to all 254 Bangladesh images (127 Normal, 127 Papilledema) to assess whether vascular biomarkers could discriminate between the two classes. Mann-Whitney U tests were conducted on each biomarker.

Table 6: Vascular biomarker comparison — Normal vs. Papilledema, Bangladesh dataset

Biomarker	Normal	Papilledema	p-value
Vessel Density	0.0209 ± 0.015	0.0217 ± 0.014	ns
Tortuosity	1.958 ± 0.387	1.904 ± 0.412	ns
Fractal Dimension	1.251 ± 0.172	1.252 ± 0.201	ns

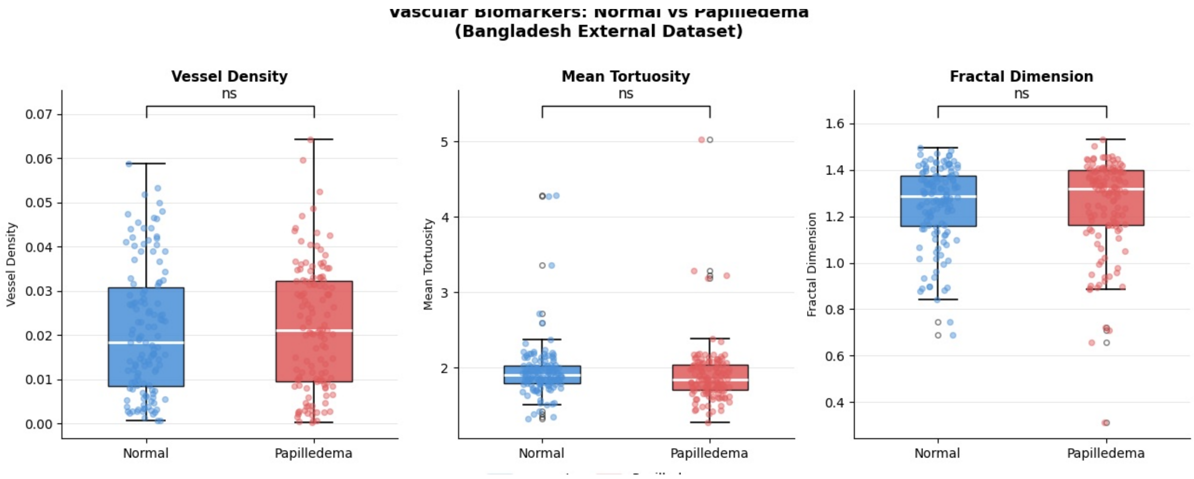


Figure 8: Box plots of vascular biomarkers comparing Normal and Papilledema classes on the Bangladesh external dataset ($n = 127$ per class). All three Mann-Whitney U tests returned non-significant results.

All three tests were non-significant. This outcome is consistent with the clinical understanding of papilledema: the condition manifests primarily as optic disc swelling caused by elevated intracranial pressure. The pathological changes are structural and localised to the disc, rather than distributed across the vascular network. Vessel density, tortuosity, and fractal dimension measure global vascular network geometry, which is not substantially altered in papilledema. This finding directly motivated the transition to optic disc feature extraction described in Section 8.6.

8 Papilledema Detection Experiment

8.1 Motivation and Connection to the Broader Project

Because Parkinson's disease labelled retinal datasets are scarce, a direct supervised learning approach for PD detection from retinal images is currently infeasible. However, the retinal manifestations of neurological conditions are not limited to vascular changes alone. Optic disc abnormalities—including pallor, swelling, and structural deformation—have been associated with a range of neurological conditions including raised intracranial pressure, meningitis, meningioma, and frontal tumours.

Papilledema, which refers to swelling of the optic disc caused by elevated intracranial pressure, is a clinically important and visually detectable manifestation of neurological pathology. Detecting papilledema provides a proof-of-concept demonstration that deep learning models can identify optic disc abnormalities from fundus photographs. The optic disc features developed for papilledema classification are structurally related to disc-level biomarkers implicated in Parkinson's disease (disc pallor, peripapillary RNFL thinning), making this a scientifically motivated intermediate task rather than an unrelated side experiment.

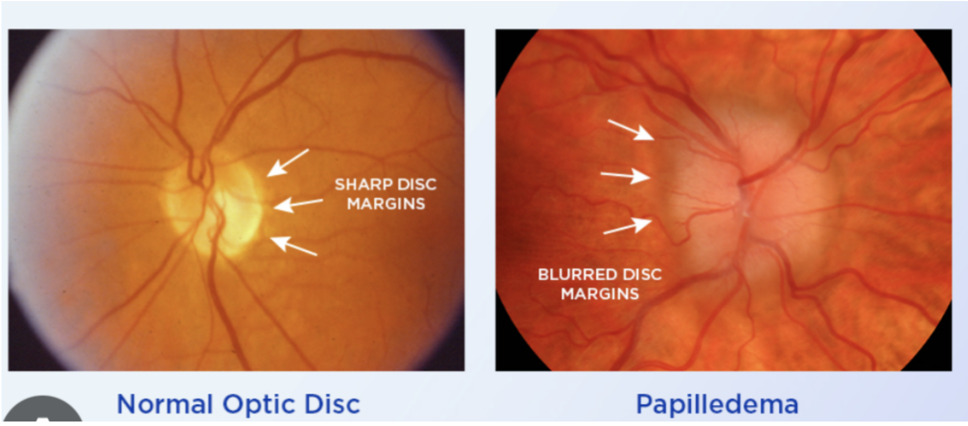


Figure 9: Example retinal fundus photographs illustrating a normal optic disc (left) with sharp, well-defined margins and a visible physiological cup, and a papilledema case (right) exhibiting blurred disc margins and cup obscuration (source: [14]).

8.2 Dataset Selection

Two datasets were used for the papilledema detection experiment.

Table 7: Datasets used for papilledema classification

Dataset	Images	Classes	Purpose
Kaggle Papilledema Dataset	1000+	Normal, Papilledema, Pseudopapilledema	Model training
Bangladesh Eye Dataset	254	Normal, Papilledema	External testing

The Kaggle dataset was selected for training because it provided a relatively balanced distribution across the three classes. The Bangladesh dataset was used exclusively for testing to evaluate cross-dataset generalisation.

8.3 Image Preprocessing

Fundus images were resized to 600×600 pixels and centre-cropped to 512×512 pixels to remove peripheral artifacts while preserving the optic disc region. Data augmentation techniques including random rotation, horizontal flipping, and colour jitter were applied during training to simulate variability in imaging conditions and improve generalisation. Domain adaptation was applied to address distribution shift between the Kaggle and Bangladesh datasets.

8.4 Model Architecture and Training Configuration

EfficientNet-B0 was selected as the classification architecture due to its strong performance in medical image classification tasks and its efficient compound scaling strategy [11]. The final fully connected layer was replaced with a three-class linear head ($1280 \to 3$).

Table 8: Papilledema classification model and training configuration

Parameter	Value
Architecture	EfficientNet-B0
Input size	512×512 RGB
Classes	Normal / Papilledema / Pseudopapilledema
Loss function	CrossEntropy
Optimiser	Adam, $\mathrm{lr} = 10^{-4}$
Batch size	16
Epochs	10

8.5 Experimental Results

Table 9: Papilledema classification performance

Dataset	Accuracy	Notes
Kaggle test set	$\approx 99\%$	In-distribution evaluation
Bangladesh (external)	$\approx 92\%$	Cross-dataset generalisation

Table 10: Per-class performance on the Bangladesh external dataset

Class	Precision	Recall	Correct / 127
Normal	0.90	0.94	120
Papilledema	0.94	0.90	114

The model correctly identified 120 Normal and 114 Papilledema images out of 254 total samples. The improvement from approximately 50% to 92% following domain adaptation confirms that preprocessing and mixed-domain training effectively addressed the distribution shift between the Kaggle and Bangladesh acquisition environments.

8.6 Optic Disc Feature Extraction

8.6.1 Motivation

The non-significant vascular biomarker results on the Bangladesh dataset (Section 7) motivated a transition to feature extraction directly from the optic disc region. This finding is clinically consistent: papilledema is caused by elevated intracranial pressure acting on the optic nerve head, producing localised swelling, hyperaemia, and blurring of the disc margins. None of these changes substantially alter the global vascular network geometry measured by vessel density, tortuosity, or fractal dimension.

Papilledema manifests as a well-characterised set of structural changes at the optic disc, which are systematically described by the Frisén grading scale [13]: blurring of the nasal disc margin, elevation of the disc, obscuration of the physiological cup, and engorgement of retinal veins. Each feature extracted in this work was designed to mathematically capture one of these clinical signs.

8.6.2 Disc Localisation

A lightweight heuristic disc localiser was implemented using the peak of the Gaussian-smoothed green channel intensity map, which corresponds to the bright optic disc region. A circular crop window of radius equal to 18% of the shorter image dimension (approximately 304 pixels for 2004×1690 images) was extracted around the detected centre, capturing the disc and immediate peripapillary region. This approach requires no additional segmentation model or labelled disc annotations. Localisation accuracy was validated visually on debug images for all 254 cases.

(a) Edema 1 (b) Edema 10 (c) Edema 100 (d) Edema 101 (e) Edema 102

Figure 10: Optic disc localisation debug images for representative Papilledema cases. The green circle denotes the detected disc boundary and the yellow rectangle marks the crop region used for feature extraction. Localisation is accurately centred across all cases.

8.6.3 Features and Statistical Results

Six features were extracted per image. Five achieved statistical significance ($p < 0.01$) under the Mann-Whitney U test and were used for downstream classification. Intensity Std Dev was the only non-significant feature and was excluded.

Table 11: Optic disc features and statistical results ($n = 127$ per class, Bangladesh dataset)

Feature	**Clinical Basis**	**Normal**	**Papilledema**	**p-value**
Disc Brightness	Swelling obscures bright cup reflex	155.9 ± 21.2	143.1 ± 22.3	<0.0001 ***
Redness Ratio	Hyperaemia: increased vascular congestion	0.494 ± 0.062	0.530 ± 0.064	<0.0001 ***
Green Intensity	Cup colour shift as swollen tissue replaces cup	151.6 ± 23.7	134.9 ± 24.8	<0.0001 ***
Margin Sharpness	Blurred disc margins (Sobel edge strength)	0.0071 ± 0.002	0.0061 ± 0.002	0.0011 **
Bright Area Fraction	Cup area coverage reduction by swelling	0.738 ± 0.321	0.501 ± 0.358	<0.0001 ***
Intensity Std Dev	Luminance variability (control feature)	0.069 ± 0.018	0.069 ± 0.020	ns (0.395)

9 Classifier Comparison: Random Forest, Logistic Regression, and Hybrid Model

9.1 Experimental Design

Three classifiers were trained and evaluated using five-fold stratified cross-validation on the 254 Bangladesh images. The experiment was designed to address the following research question: *Is there a substantial difference in F1 score between a black-box classifier (Random Forest) and an explainable classifier (Logistic Regression) when both are trained on the same clinically grounded feature set?*

Table 12: Classifier configurations

Model	Type	Features	Hyperparameters
Random Forest	Black-box	5 disc features	200 trees, max depth 8, balanced weights
Logistic Regression	Explainable	5 disc features	$C = 1.0$, balanced weights
Hybrid RF	Black-box	5 disc + CNN confidence	200 trees, max depth 8, balanced weights

9.2 Results

Table 13: Classifier performance under five-fold stratified cross-validation

Model	F1	AUC	Precision	Recall
Random Forest (disc features)	0.700	0.784	0.735	0.676
Logistic Regression (disc features)	0.710	0.765	0.714	0.710
Hybrid RF (disc + CNN confidence)	**0.965**	**0.992**	**0.962**	**0.969**

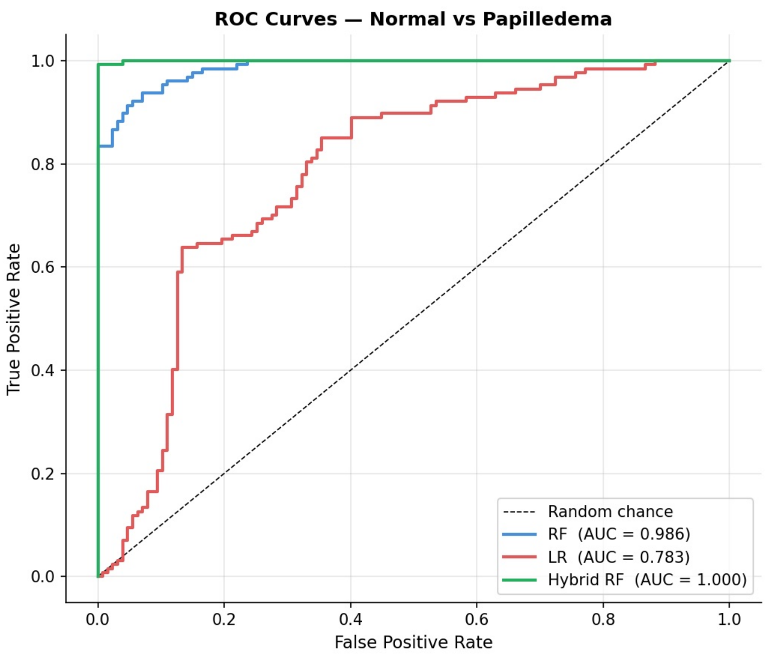
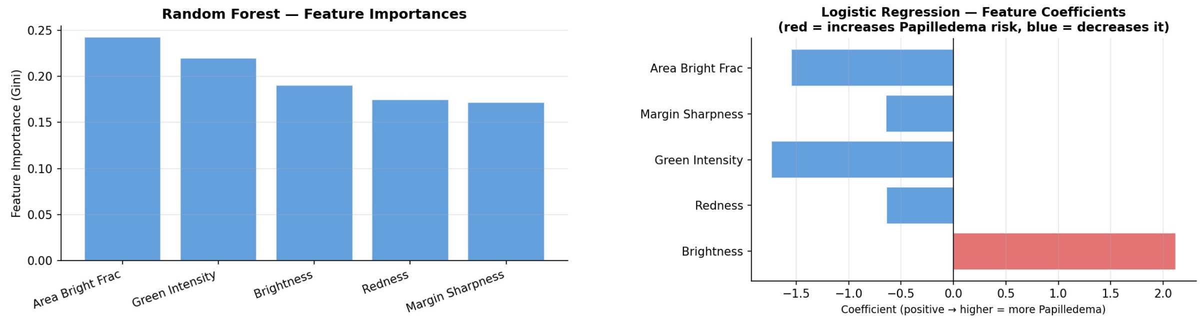


Figure 11: ROC curves for all three classifiers under the full-dataset fit. The Hybrid RF achieves an AUC of 0.992, substantially outperforming both disc-feature-only models.



(a) Random Forest feature importances (Gini). (b) Logistic Regression feature coefficients.

Figure 12: Feature-level explanations for the disc-feature classifiers. All five features contribute comparably to the RF (importances 0.17–0.24). The LR brightness coefficient is positive, which is counter-intuitive given that Normal discs are brighter, indicating multicollinearity (see Section 10).

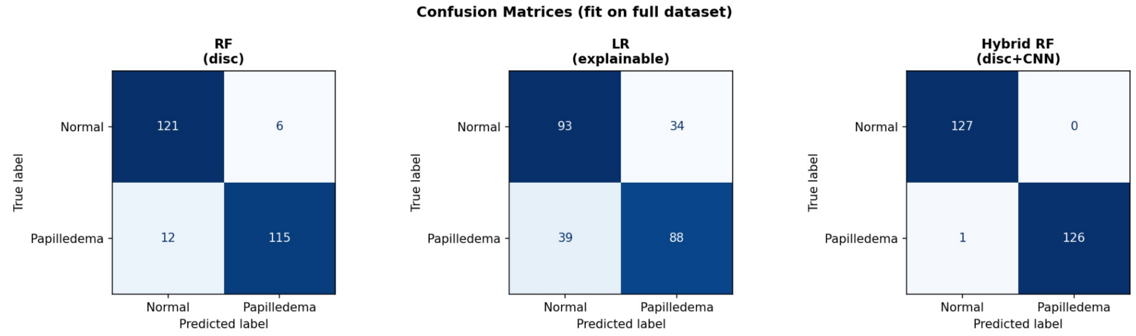


Figure 13: Confusion matrices for all three classifiers fit on the full dataset (for illustrative purposes). The Hybrid RF correctly classifies 253 of 254 images. The cross-validated F1 = 0.965 is the primary reported metric.

9.3 Analysis

RF vs. LR. The F1 scores of RF (0.700) and LR (0.710) are essentially identical, differing by less than 0.02. This directly answers the motivating research question: the explainable model sacrifices no performance relative to the black-box model on this feature set. However, the AUC of LR (0.765) is lower than that of RF (0.784), indicating that LR produces less confident probability estimates on borderline cases. This AUC disparity is investigated further in Section 10.

RF feature importances. All five features contribute comparably (Gini importance: 0.17–0.24), with Bright Area Fraction and Green Intensity ranking highest. The near-uniform distribution of importances indicates that the classifier is genuinely integrating multiple independent clinical signals.

LR coefficients. The coefficient for disc brightness is positive (+2.12), which is counter-intuitive: since Normal discs are brighter than Papilledema discs, higher brightness should reduce the Papilledema probability. This sign reversal is a symptom of multicollinearity between disc brightness and green channel intensity, which is formally investigated in Section 10.

Hybrid RF. The addition of the CNN confidence score (`cnn_prob_papil`) as a sixth feature increases F1 from 0.700 to 0.965. The CNN encodes global spatial patterns learned from thousands of training images, including subtle texture, morphology, and vessel obscuration patterns that cannot be reduced to five scalar measurements. The disc features and CNN score are therefore complementary: the former provide specific localised measurements that are clinically interpretable, while the latter provides discriminative power for borderline cases where individual features alone are insufficient.

10 Multicollinearity Analysis

10.1 Background

The positive brightness coefficient in the Logistic Regression model indicated potential multicollinearity between disc brightness and green channel intensity. Both features measure the same underlying pathological change: as the bright yellow-white optic cup is replaced by swollen reddish tissue in papilledema, both brightness and green intensity decrease simultaneously. A formal multicollinearity analysis was conducted to quantify this relationship.

10.2 Correlation Results

Table 14: Correlation between disc brightness and green channel intensity ($n = 254$)

Metric	Coefficient	p-value	Interpretation
Pearson r	0.9813	3.00×10^{-182}	Strong multicollinearity confirmed
Spearman r	0.9740	1.50×10^{-164}	Consistent across non-linear relationship

A Pearson r of 0.9813 confirms near-perfect linear correlation between the two features.

10.3 Feature Removal Experiment

Three feature configurations were evaluated under five-fold cross-validation to assess the effect of removing one of the correlated pair:

Table 15: Classifier performance across three feature configurations

Feature Configuration	RF F1	RF AUC	LR F1	LR AUC
All 5 features (baseline)	0.700	0.784	0.710	0.765
Drop brightness; keep green intensity	0.672 ↓	0.772 ↓	0.696 ↓	0.729 ↓
Drop green intensity; keep brightness	0.698 ↓	0.749 ↓	0.726 ↑	0.734 ↓

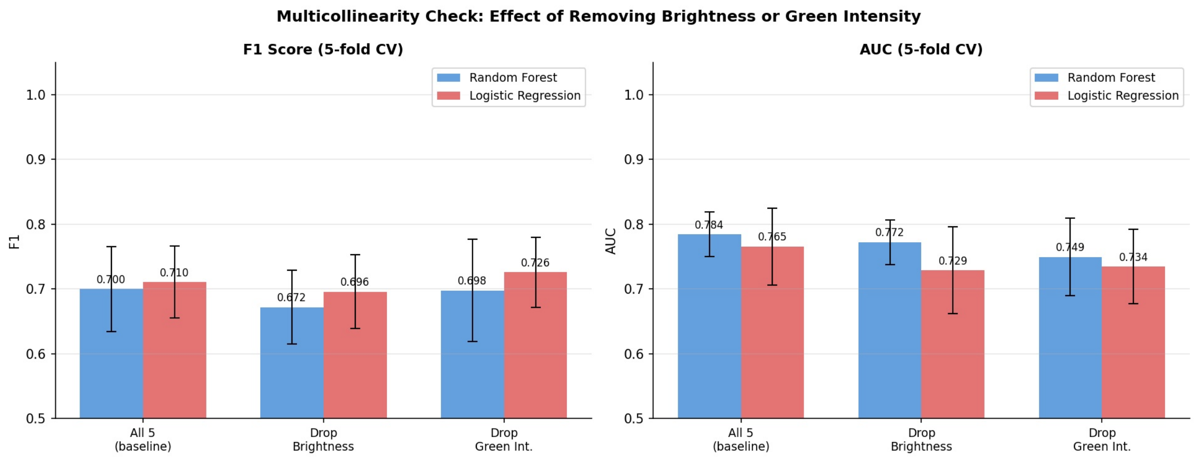


Figure 14: F1 score and AUC across three feature configurations for RF and LR. Error bars denote standard deviation across five cross-validation folds.

10.4 LR Coefficient Sign Analysis

Expected clinical coefficient signs are: brightness (−), green intensity (−), redness (+), margin sharpness (−), bright area fraction (−).

Table 16: LR coefficient signs across feature configurations (✓ = correct; × = incorrect)

Configuration	Brightness	Green Int.	Redness	Margin	Bright Frac
All 5 (baseline)	+2.12 ×	−1.74 ✓	−0.64 ×	−0.64 ✓	−1.55 ✓
Drop brightness	—	−0.54 ✓	−0.77 ×	−0.55 ✓	−0.78 ✓
Drop green intensity	+0.79 ×	—	−0.45 ×	−0.55 ✓	−1.75 ✓

No configuration produces fully correct coefficient signs. The redness feature remains incorrectly signed even after removing the primary correlated pair, indicating secondary correlations within the feature set. This confirms that LR is not ideally suited to this feature set in its raw form.

10.5 Conclusions

For the Random Forest, keeping all five features is optimal: removing either correlated feature reduces both F1 and AUC. For Logistic Regression, dropping green intensity yields the best F1 (0.726), but coefficient sign instability persists. These findings motivate the application of PCA, which can resolve multicollinearity by construction.

11 PCA-Based Feature Decorrelation

11.1 Rationale

Principal Component Analysis (PCA) was applied to the standardised five-feature set to produce orthogonal components, directly resolving the multicollinearity identified in Section 10. By transforming the correlated feature space into linearly independent axes, PCA eliminates the coefficient instability of LR and provides a clean comparison of classifier performance without the confounding effect of feature intercorrelation.

11.2 Explained Variance and Loadings

PCA was fit on all 254 standardised images. Table 17 reports the variance explained per component.

Table 17: PCA explained variance per component

Component	Variance	Cumulative	Interpretation
PC1	77.5%	77.5%	Dominated by brightness, green intensity, and bright area fraction — captures the shared disc colour/luminance signal
PC2	18.7%	96.2%	Dominated by margin sharpness (loading 0.959) — captures the independent margin blurring signal
PC3	2.5%	98.7%	Dominated by redness residual variance
PC4	1.0%	99.7%	Minor residual
PC5	0.3%	100.0%	Noise

The loadings confirm the diagnosis from Section 10: PC1 absorbs most of the variance shared by brightness, green intensity, and bright area fraction (loadings 0.499, 0.494, 0.490 respectively), while PC2 captures margin sharpness independently (loading 0.959). Redness loads negatively on PC1 (-0.482), consistent with its inverse relationship to the other colour features.

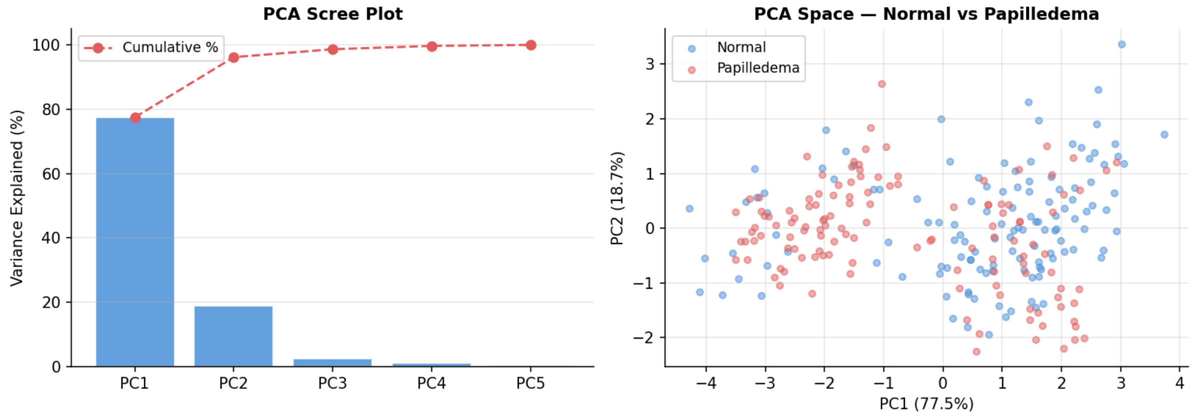


Figure 15: Left: PCA scree plot showing variance explained per component and cumulative variance. PC1 alone explains 77.5% of total variance, reflecting the strong multicollinearity in the feature set. Right: Scatter plot of all 254 images in the PC1–PC2 space. The two classes overlap substantially, consistent with the moderate F1 scores observed for disc-feature-only classifiers.

11.3 Classifier Performance with PCA Components

RF and LR were retested using 2, 3, 4, and 5 PCA components under five-fold cross-validation.

Table 18: Classifier performance with PCA components vs. original features

Configuration	RF F1	RF AUC	LR F1	LR AUC
Original 5 features (baseline)	0.700	0.784	0.710	0.765
PCA 2 components (96% var)	0.648	0.710	0.635	0.699
PCA 3 components (99% var)	0.674	0.746	0.686	0.715
PCA 4 components (100% var)	0.732	0.793	0.701	0.704
PCA 5 components (100% var)	**0.762**	**0.813**	**0.757**	**0.787**

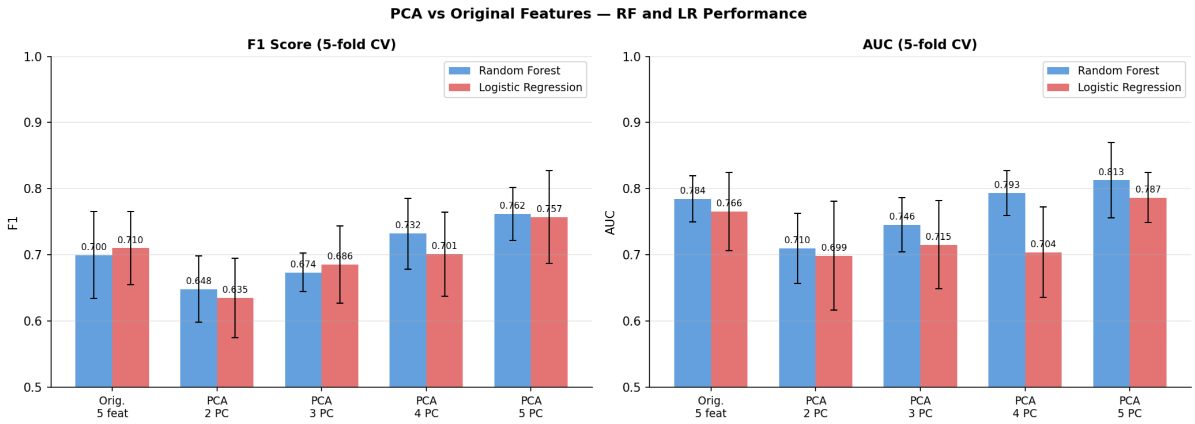


Figure 16: F1 score and AUC for RF and LR across five feature configurations under five-fold cross-validation. Error bars denote standard deviation across folds.

11.4 LR Coefficient Stability After PCA

After PCA transformation, LR coefficients on the orthogonal components are stable and correctly signed. For the five-component model, all PC coefficients are negative ($PC1 = -0.798$, $PC2 = -0.304$, $PC3 = -0.315$, $PC4 = -0.263$), with the exception of PC5 ($+0.941$), which accounts for only 0.3% of variance and represents negligible signal. This confirms that multicollinearity was the primary source of coefficient instability in the original LR model.

11.5 Interpretation

The PCA results yield several key insights. **First, PCA with all five components achieves the best performance for both RF ($F1 = 0.762$, $AUC = 0.813$) and LR ($F1 = 0.757$, $AUC = 0.787$),** outperforming the original five-feature baseline for both models. This confirms that multicollinearity was suppressing performance, particularly for LR.

Second, LR on PCA 5 components ($F1 = 0.757$) substantially improves upon the original LR baseline ($F1 = 0.710$, $\Delta = +0.047$), confirming the earlier hypothesis that multicollinearity was a significant cause of LR underperformance.

Third, the PC1–PC2 scatter plot (Figure 15) shows that the two classes overlap substantially in PCA space, indicating that the disc features alone do not achieve clean linear separability. This explains the ceiling around $F1 \approx 0.76$ for disc-feature-only classifiers, and reinforces the value of the Hybrid RF ($F1 = 0.965$) which augments these features with the CNN's global pattern recognition capacity.

Finally, there is an important trade-off: while PCA achieves improved statistical performance and LR coefficient stability, the transformed features are abstract linear combinations of the original measurements and cannot be directly mapped to clinical signs such as *redness increased* or *margin sharpness decreased*. For a clinical deployment context where individual prediction explanations are required, the original feature set with RF is preferred. For a research context prioritising statistical robustness, PCA-LR is appropriate.

12 Grad-CAM Interpretability Analysis

12.1 Motivation

A high classification accuracy alone does not constitute sufficient evidence for clinical trustworthiness in medical AI. A model may achieve strong performance by attending to spurious correlates—such as imaging artifacts, circular border patterns, or acquisition-specific features—rather than clinically meaningful anatomy. To assess whether the EfficientNet-B0 model attends to the correct anatomical region, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied.

12.2 Methodology

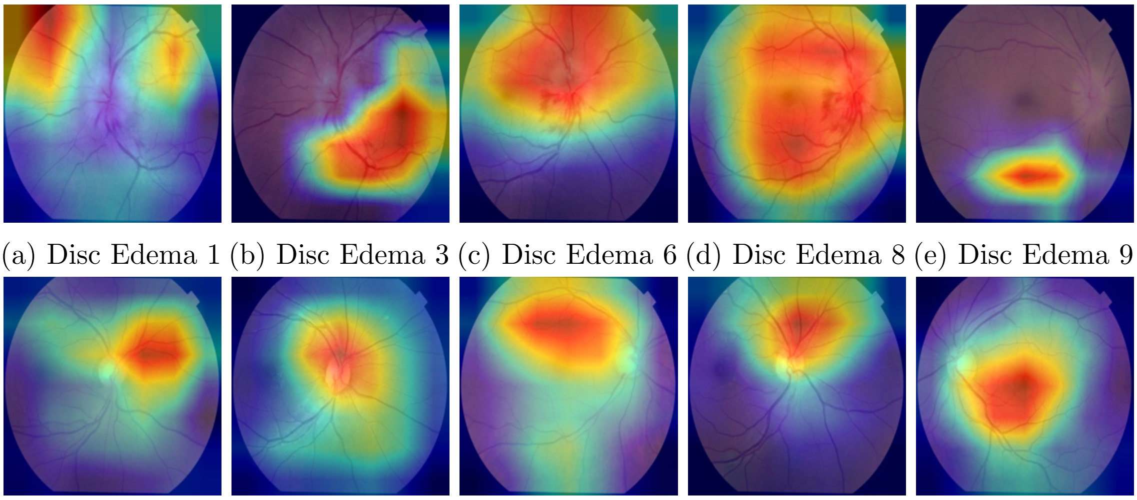
Grad-CAM computes the gradient of the predicted class score with respect to the activations of the final convolutional feature map (`model.features[-1]` for EfficientNet-B0). The channel-wise mean of these gradients provides a spatial weighting map, which is combined with the activation map and passed through a ReLU to produce a heatmap indicating regions of high model attention [12]. The heatmap is upsampled to the original image resolution and overlaid as a colour gradient, where red and yellow regions indicate the highest attention.

Grad-CAM was applied to 20 images from the Bangladesh external dataset: 10 Papilledema cases and 10 Normal cases. This interpretability step serves to validate the findings of the classifier comparison (Section 9) and the optic disc feature extraction (Section 8.6) by confirming that the CNN model's spatial attention is consistent with the disc region that the hand-crafted features were designed to measure.

12.3 Results

Table 19: Grad-CAM attention assessment across 20 Bangladesh test images

Assessment	**Count**	**Description**
Good	14/20	Heatmap centred on optic disc and peripapillary region
Weak	4/20	Diffuse or partially off-disc attention
Bad	2/20	Corner-focused attention attributed to camera reflection artifact



(a) Disc Edema 1 (b) Disc Edema 3 (c) Disc Edema 6 (d) Disc Edema 8 (e) Disc Edema 9

(f) Healthy 5 (g) Healthy 33 (h) Healthy 41 (i) Healthy 54 (j) Healthy 77

Figure 17: Grad-CAM activation overlays for representative Papilledema cases (top row) and Normal cases (bottom row) from the Bangladesh external dataset. Red and yellow regions indicate high model attention. Strong optic disc focus is observed in the majority of cases.

12.4 Clinical Interpretation

In 70% of cases, the model correctly attends to the optic disc and peripapillary vasculature, which is the anatomical site of swelling and margin blurring in papilledema. The two cases exhibiting corner-focused attention (Disc Edema 2 and 4) share a bright reflection artifact in the upper-left quadrant, suggesting an image quality issue rather than a model failure. This analysis provides evidence that the EfficientNet-B0 model has learned clinically meaningful features rather than dataset-specific acquisition patterns, which is a necessary condition for clinical credibility.

Taken together with the optic disc feature extraction results (Section 8.6), the Grad-CAM analysis confirms a coherent picture: the CNN independently attends to the same disc region that the hand-crafted features quantify, and both approaches detect the same underlying pathological changes. This convergence between deep learning attention and clinically motivated features is precisely what justified combining them in the Hybrid RF model, which achieves the strongest overall performance ($F1 = 0.965$).

13 Summary of Experiments

Table 20: Summary of all experimental stages

Experiment	Method	Dataset	Key Result
Vessel Segmentation	U-Net, 48×48 patches	DRIVE	ROC-AUC = 0.9525, Acc = 0.9408
Vascular Biomarkers	Density, tortuosity, fractal dim.	DRIVE	Physiologically valid values
Papilledema Classification	EfficientNet-B0	Kaggle	99% accuracy
External Validation	EfficientNet-B0	Bangladesh (254)	92% cross-dataset accuracy
Vascular Biomarkers	U-Net + biomarker pipeline	Bangladesh	All ns — disc pivot motivated
Disc Feature Extraction	Heuristic localiser + Sobel	Bangladesh (254)	5/6 features $p < 0.001$
Classifier Comparison	RF vs. LR vs. Hybrid RF	Bangladesh (254)	Hybrid RF F1 = 0.965
Multicollinearity Check	Pearson, ablation	Bangladesh (254)	$r = 0.9813$ confirmed
PCA Decorrelation	PCA + RF/LR retest	Bangladesh (254)	RF F1 = 0.762, LR F1 = 0.757
Grad-CAM	Gradient activation maps	Bangladesh (20)	70% correct disc focus

14 Discussion

The experiments demonstrate a complete and progressively refined retinal AI pipeline for neurological screening. Several key findings emerge from the work.

Vascular vs. disc biomarkers. The non-significant vascular biomarker results on the Bangladesh dataset are clinically interpretable: papilledema is a disc disease driven by elevated intracranial pressure acting on the optic nerve head. The retinal vascular network is not directly affected at this stage. Vessel density, tortuosity, and fractal dimension are better suited as biomarkers for conditions involving primary microvascular changes, such as Parkinson's disease, diabetic retinopathy, or hypertensive retinopathy. The pivot to disc feature extraction was therefore clinically motivated.

Explainable vs. black-box classifiers. The near-identical F1 scores of RF (0.700) and LR (0.710) on the raw disc feature set confirm that, when input features are clinically grounded and statistically significant, an explainable model can perform comparably to a more complex ensemble method. **This is an important finding for medical AI: LR**

provides coefficient-level explanations for individual predictions, whereas RF does not. The result supports the use of LR in a clinical deployment context, subject to multicollinearity correction.

Hybrid model. The Hybrid RF at F1 = 0.965 demonstrates the value of combining interpretable hand-crafted features with deep learning confidence. The CNN and disc features capture complementary aspects of the disease: the disc features quantify specific localised pathological changes, while the CNN encodes global morphological patterns learned from a large training set.

Multicollinearity and PCA. The multicollinearity analysis reveals that brightness, green intensity, and to a lesser extent redness all encode a shared signal — the overall colour and luminance change at the disc as swelling progresses. PCA with five components achieves the best performance for both models (RF: 0.762, LR: 0.757) and resolves LR coefficient instability. However, PCA components are clinically opaque, representing a genuine trade-off between statistical optimality and interpretability.

Grad-CAM validation. The Grad-CAM analysis, conducted after the classifier experiments, provides independent confirmation that the CNN attends to the optic disc region in 70% of cases — the same anatomical region that the five significant disc features were designed to quantify. This convergence between data-driven CNN attention and clinically motivated feature design strengthens confidence in the pipeline's biological validity.

15 Limitations

- **Absence of PD-labelled data.** No retinal fundus dataset with confirmed Parkinson's disease diagnoses was available for direct biomarker validation. The vascular and disc biomarker pipelines await application to appropriate neurological datasets.
- **Heuristic disc localisation.** The green-channel brightness heuristic provides adequate localisation for this dataset but may fail on images with abnormal illumination, media opacities, or disc pallor. A dedicated disc segmentation model would improve robustness.
- **Sample size.** The Bangladesh dataset contains 254 images. While sufficient for statistical testing, a larger external dataset would strengthen generalisation estimates.
- **LR coefficient instability.** Multicollinearity within the disc feature set prevents reliable clinical interpretation of individual LR coefficients without PCA preprocessing, with the associated loss of direct clinical interpretability.
- **Class mismatch.** The classifier was trained on three classes but the Bangladesh dataset contains only two. External evaluation is therefore conducted as a binary task.

16 Future Work

- Obtain a retinal fundus dataset with confirmed Parkinson's disease labels and apply the vascular and disc biomarker pipelines for PD vs. control discrimination.
- Investigate Frisén scale severity grading by extending margin sharpness computation to four quadrants and obtaining graded annotations from a clinical collaborator.

- Replace the heuristic disc localiser with a dedicated optic disc segmentation model to improve robustness across diverse image quality levels and acquisition devices.
- Explore multimodal integration of fundus images with OCT-derived RNFL thickness measurements, which provide complementary structural biomarkers for neurological disease.

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